

Maulana Abul Kalam Azad University of Technology, West Bengal
(Formerly known as West Bengal University of Technology)
Syllabus of *B.Sc. in Data Science*
Effective from academic session 2023-2024

Paper Code: BDS302 & BDS392

Paper Name: *Artificial Intelligence & Machine Learning - 1 & AI ML-1 Lab*

Credits: 5 (3Th+2P)

Module 1: Introduction to Artificial Intelligence

- Definition and scope of Artificial Intelligence
- Historical overview and milestones in AI
- AI applications in various fields
- Agents and environments

Module 2: Problem Solving and Searching

- Problem-solving agents
- State space representation
- Uninformed search algorithms (breadth-first search, depth-first search)
- Informed search algorithms (heuristic search, A* algorithm)
- Appropriate laboratory work in Python/R

Module 3: Knowledge Representation and Reasoning

- Propositional logic
- First-order logic and predicate calculus
- Knowledge representation using logic
- Inference rules and resolution

Module 4: Expert Systems

- Introduction to expert systems
- Knowledge engineering process
- Rule-based systems
- Forward chaining and backward chaining

Module 5: Introduction to Statistical Learning

- Overview of statistical learning and its applications
- Supervised vs. unsupervised learning
- Bias-variance tradeoff
- Performance evaluation and cross-validation

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Module 6: Linear Regression

- Simple linear regression
- Multiple linear regression
- Model assessment and selection
- Polynomial regression and model flexibility
- Appropriate laboratory work in Python/R

Module 7: Classification

- Logistic regression
- Linear discriminant analysis (LDA)
- K-nearest neighbors (KNN) classification
- Model evaluation and validation techniques
- Appropriate laboratory work in Python/R

Module 8: Resampling Methods

- Cross-validation
- Bootstrap methods for estimating standard errors
- Model selection using cross-validation and bootstrapping
- Appropriate laboratory work in Python/R

Module 9: Dimensionality Reduction

- Principal Component Analysis (PCA)
- Singular Value Decomposition (SVD)
- Applications of dimensionality reduction
- Appropriate laboratory work in Python/R

Module 10: Introduction to Machine Learning

- Overview of different machine learning algorithms
- Decision Trees and Random Forests
- Support Vector Machines (SVM)
- Ensemble methods (bagging and boosting)
- Appropriate laboratory work in Python/R

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Module 11: Unsupervised Learning

- K-means clustering
- Hierarchical clustering
- Association rules and market basket analysis
- Anomaly detection
- Appropriate laboratory work in Python/R

References:

1. Dan W. Patterson (1990). *Introduction to artificial intelligence and expert systems*. Englewood Cliffs, N.J : Prentice Hall
2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning with applications in R (2nd ed.)*. Springer.
3. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An introduction to statistical learning with applications in Python (1st ed.)*. Springer.